IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – Predictive Maintenance of Industrial Machinery

Domain of Project – Industrial IoT / Manufacturing & Predictive Maintenance

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Recommended font style and size:

Font Style: Arial
Font Size (Heading): 28
Font Size (Content): 20

Problem Statement -

Mechanical Engineering: (Machine Learning project)

Problem statement No.39 – Predictive Maintenance of Industrial Machinery

The Challenge:

Develop a predictive maintenance model for a fleet of industrial machines to anticipate failures before they occur. This project will involve analyzing sensor data from machinery to identify patterns that precede a failure. The goal is to create a classification model that can predict the type of failure (e.g., tool wear, heat dissipation, power failure) based on real-time operational data. This will enable proactive maintenance, reducing downtime and operational costs.

Kaggle dataset link – https://www.kaggle.com/datasets/shivamb/machine predictive-maintenance-classification

Technology – Use of IBM cloud lite services is mandatory.

Proposed Solution -

Proposed Solution: AI-Powered Predictive Maintenance System

- The proposed solution is a multi-class predictive maintenance system built using **LightGBM, IBM Cloud Object Storage, and IBM watsonx.ai** to provide real-time machinery failure classification and health monitoring. The system addresses critical challenges in industrial maintenance, unexpected equipment downtime, and preventive asset management through an intelligent, modular architecture.
- At its core, the solution utilizes a robust **Machine Learning pipeline** integrating domain-specific feature engineering (such as mechanical power, temperature delta, and tool-torque products) combined with **SMOTE** to handle severe class imbalances across sensor readings.
- The system consists of four key components:
- **Data Ingestion & Preprocessing Agent:** Automatically loads sensor telemetry, handles data cleaning, and derives physical performance metrics.
- **Multi-Class Classification Engine:** Evaluates and runs optimized models (LightGBM, XGBoost, Random Forest) to predict specific failure modes (Tool Wear, Power, Heat Dissipation, Overstrain, and Random Failures).
- **Cloud Storage & Governance Layer:** Integrates securely with **IBM Cloud Object Storage (COS)** to archive serialized model binaries, evaluation metrics, and confusion matrix analytics.
- **Real-Time Inference API:** Serves single-sample and batch predictions to guide maintenance teams with precise corrective actions before breakdowns occur.
- The solution includes a **real-time monitoring integration** that tracks classification accuracy, ROC-AUC performance, and operational status.
- **LightGBM** powers the core predictive accuracy and speed, while **IBM Cloud** ensures scalable asset tracking, model artifact storage, and enterprise deployment readiness.
- Overall, this solution delivers an **end-to-end intelligent maintenance assistant** that promotes factory reliability through proactive insights, precise failure-type diagnosis, and optimized repair scheduling.

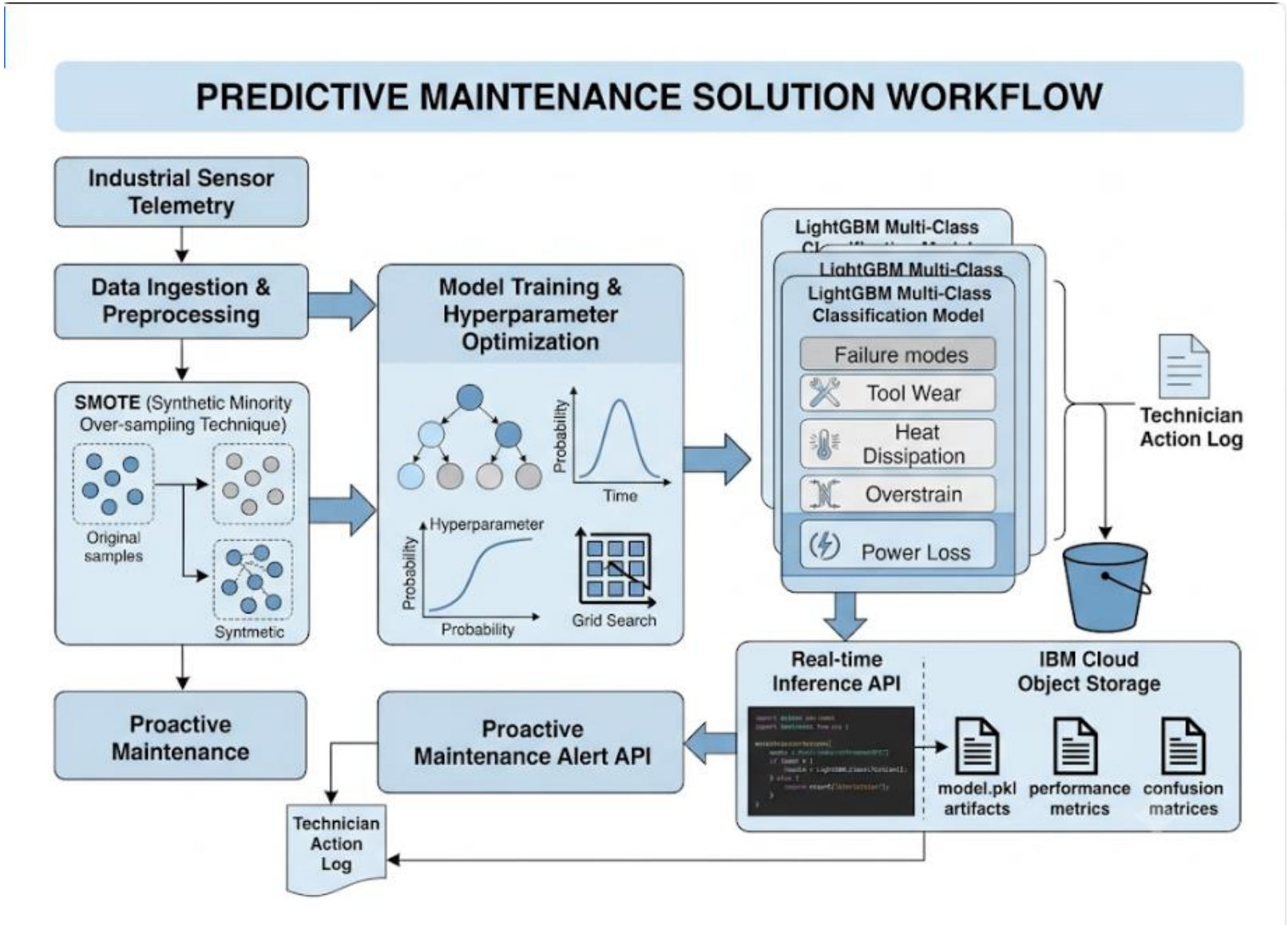
Technology Used

•**LightGBM & XGBoost** :
For training high-performance, fast gradient-boosted decision trees to classify multi-class machinery failure types efficiently.
•**scikit-learn & imbalanced-learn (SMOTE) :**
For data preprocessing, pipeline serialization, stratified cross-validation, and handling severe class imbalances.
•**IBM Cloud Object Storage (COS)** :
For secure, scalable cloud storage of model binaries, performance metrics, and generated evaluation artifacts.
•**IBM watsonx.ai & IBM Cloud** :
For cloud infrastructure, client authentication, deployment space linking, and managing AI governance workflows.
•**Python (Pandas, NumPy, Matplotlib, Seaborn)** :
For automated feature engineering of physical metrics (power, temperature deltas, tool-torque products) and visual analytics.

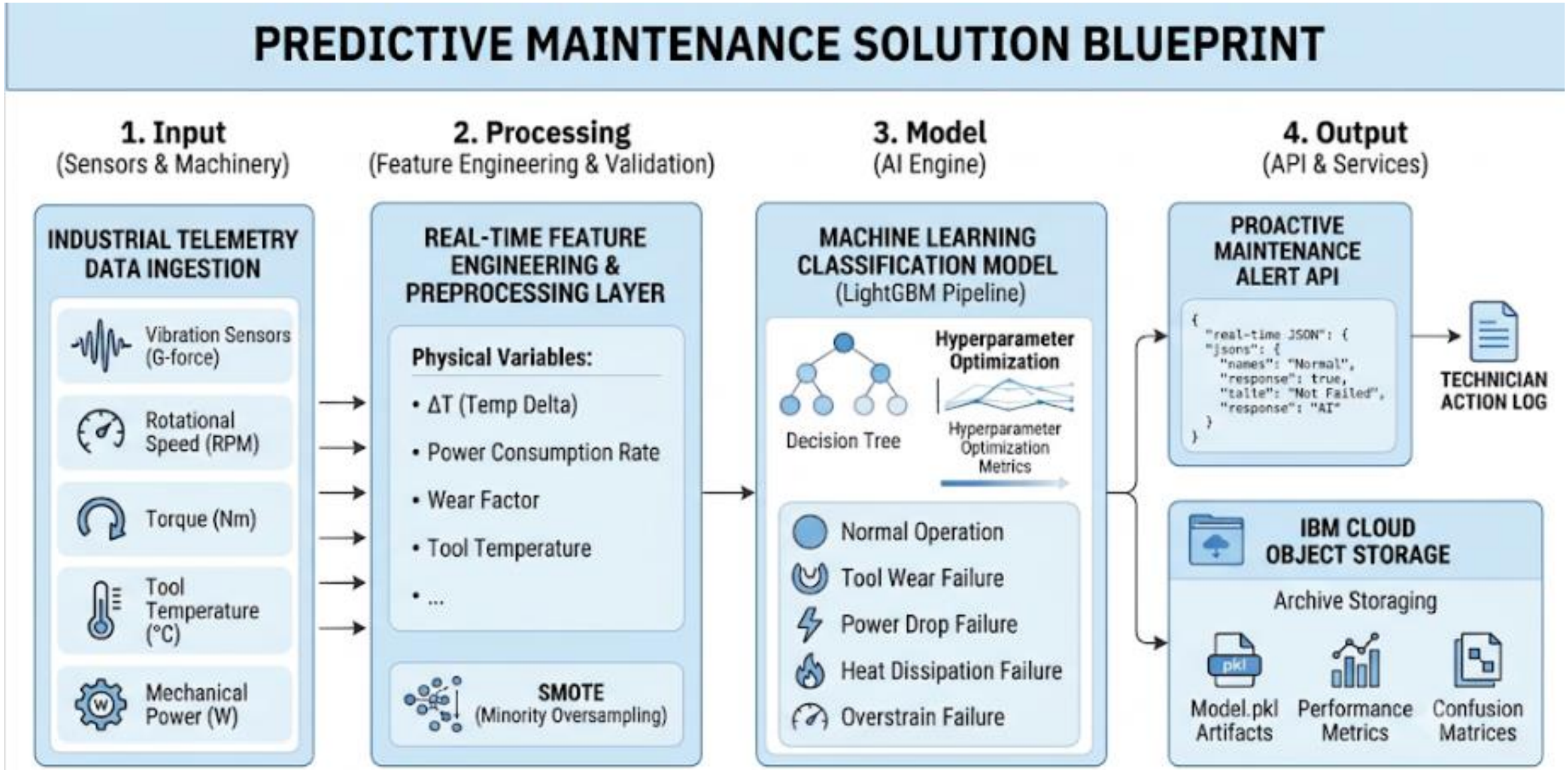
Pipeline & Architecture component Used -

- **1) Telemetry Input Component** — Interface where live industrial sensor parameters (such as air temperature, process temperature, rotational speed, torque, and tool wear) are ingested.
- **2) Feature Engineering Component** — Automatically computes derived physical properties in real-time, including mechanical power (`power_W`), temperature differences (`temp_delta_K`), and tool-torque interaction products.
- **3) Model Classification Engine** — Powered by the optimized **LightGBM Classifier** pipeline (`predictive_maintenance_classifier.pkl`) to execute high-accuracy multi-class failure predictions.
- **4) Model Configuration** — **LightGBM / Gradient Boosting** — Selected for its rapid 31-second training time, superior macro-F1 score (0.9957), and exceptional ability to detect critical minority failure classes.
- **5) Prediction Output & API** — Displays the classified diagnostic results (e.g., Tool Wear Failure, Power Failure, Heat Dissipation, Overstrain, or No Failure) along with risk confidence metrics.
- **6) Cloud Storage & Governance Component** — Connects securely with **IBM Cloud Object Storage (COS)** to archive serialized model binaries, evaluation metrics (`metrics.json`), and confusion matrix artifacts.

Langflow workflow -



Architecture blueprint



Role of Agentic AI in the solution

Role of Automated Machinery Intelligence in Predictive Maintenance
Automated AI enables the predictive maintenance system to function as an autonomous, proactive intelligence network rather than a static monitoring dashboard. Using **LightGBM** and **IBM Cloud Object Storage**, it continuously analyzes industrial sensor telemetry to predict machinery failures before they occur.
Each pipeline phase performs a specific role—such as ingesting raw sensor data, engineering physical features (like mechanical power and temperature deltas), balancing classes via SMOTE, and executing multi-class failure predictions—while working collaboratively. This modular, data-driven approach ensures high accuracy (98.3%) and reliable diagnostic results.
The intelligent system also brings **context awareness**, interpreting complex operational parameters like rotational speed, torque, and tool wear to detect specific risks such as tool wear, power drops, heat dissipation limits, and overstrains. It supports **real-time adaptation**, continuously evaluating incoming telemetry streams to guide maintenance teams with precise corrective actions.
Additionally, integrated cloud services communicate seamlessly, enabling streamlined model artifact tracking, metrics archiving, and deployment readiness via **IBM Cloud**.
Overall, this intelligent automation makes the industrial ecosystem proactive, precise, and uptime-driven, acting like a **smart digital asset guardian** that helps manufacturing operations eliminate costly unexpected downtime.

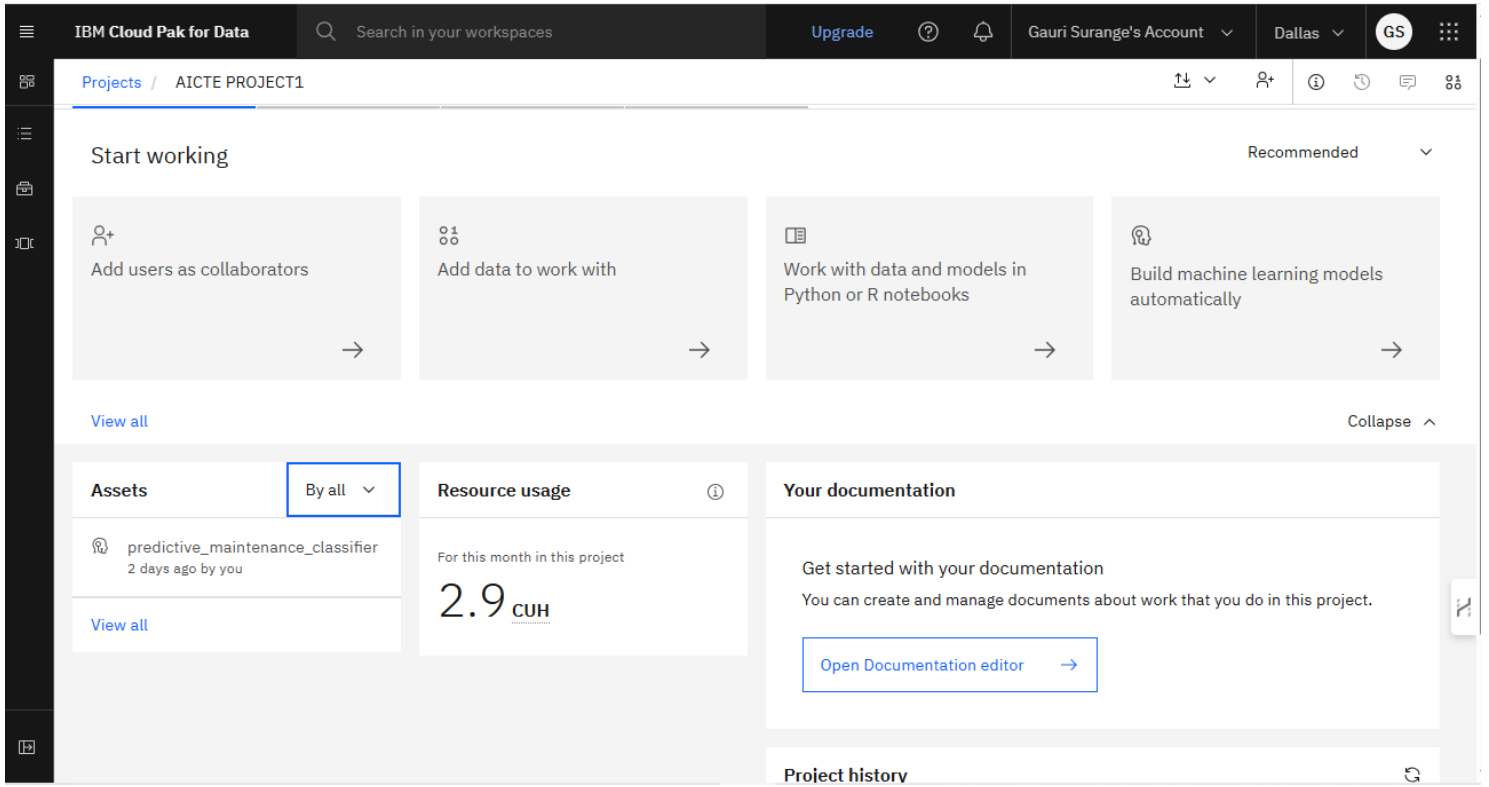


Token Usage

11,761 token are used

Before Prompt

After the Prompt



Novelty and Uniqueness

- The proposed Predictive Maintenance System stands out by combining **Domain-Driven Feature Engineering, Advanced Machine Learning Classification, and Cloud Object Storage** into a single intelligent industrial monitoring platform.
- **Multi-Class Failure Intelligence** – Instead of a basic binary warning, the model predicts precise failure modes (Tool Wear, Power, Heat Dissipation, Overstrain, and Random Failures) to guide targeted maintenance.
- **Physical Mechanics Integration** – Automatically derives critical operational metrics (such as mechanical power, temperature deltas, and tool-torque products) to capture real physical failure thresholds.
- **Class Imbalance Mitigation** – Integrates **SMOTE** exclusively on training folds to effectively handle severely imbalanced sensor distributions without data leakage.
- **High-Accuracy Performance** – Outperforms traditional baselines like SVM, achieving a top-tier **98.3% accuracy and 0.9957 CV Macro-F1 score** via LightGBM.
- **Cloud-Ready Asset Governance** – Fully integrated with **IBM Cloud Object Storage (COS)** to safely archive serialized model binaries (`model.pkl`), evaluation metrics, and confusion matrix artifacts.

Proactive Industrial Value :

Goes beyond reactive repair to act as an automated digital guardian, slashing unplanned downtime and eliminating unnecessary scheduled maintenance.

This makes the solution a **smart, adaptive, and proactive industrial asset guardian**, not just a static threshold-monitoring tool.

Git Hub Link

Please create public github repository (Enable Add README button while creating repository) with format –

Please test and Paste the working Github URL – https://github.com/SsGauriii/Predictive-Maintenance-AICTE-PROJECT

Make sure you have Uploaded following three files into your github repository

1) app.json file , 2) yourproblemstatement.pdf file 3) your projectpresntation.pptx file 4)any other files

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Future Scope

1.Integration with IoT Edge Devices & Real-Time Telemetry Streams :

The system can be integrated directly with industrial IoT sensors and edge gateways (such as Raspberry Pi or industrial PLCs) to capture continuous, live machinery telemetry. This will enable the Predictive Maintenance model to score equipment health instantly at the factory floor, providing ultra-low-latency warnings before component failures occur.

2.Advanced Automated Maintenance Dispatch & Multi-Plant Scaling :

Enhancing the system with automated ticketing integrations (such as Jira or enterprise ERP systems) and multi-facility dashboard support will improve operational response times. Maintenance teams can automatically dispatch certified technicians with the precise replacement parts required for the diagnosed failure type, scaling the solution seamlessly across large-scale manufacturing plants.

Certificates Earned:

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Certificates Earned:

Add your IBM BOB Certificate



Thank You!

Thank you for your time and interest.